Mapping Marine Oil Spill Concentrations From SAR Images Using a Co-Polarization Difference-Based Method

Honglei Zheng, *Senior Member, IEEE*, Yunhua Wang, Peng Ren, *Senior Member, IEEE*, and Weimin Huang, *Senior Member, IEEE*

***Abstract*—Accurate mapping of oil concentrations is essential for effective response to oil spill emergencies. The complexity of microwave scattering over oil-contaminated sea surfaces poses substantial challenges for synthetic aperture radar (SAR) applications, primarily due to the limited understanding of non-Bragg scattering mechanisms. This knowledge gap restricts the development of robust retrieval algorithms for quantifying oil spill concentrations. To address this issue, a novel retrieval approach is proposed based on the co-polarization difference (PD), which is independent of non-Bragg scattering. The influence of oil on the sea surface is attributed to two dominant factors: suppression of short gravity-capillary waves and reduction in the effective dielectric constant. By analyzing SAR imagery of oil spills with varying concentrations, it is found that the damping effect of oil spills on small-scale waves can be predicted using the Marangoni damping model. Once the contribution of wave suppression to PD reduction is isolated, the residual PD variation is attributed to changes in the dielectric constant. Oil concentration is then retrieved by comparing the PD of each pixel within the contaminated area to a theoretical PD lookup table. The proposed method is validated using simulated SAR datasets representing different oil concentrations and subsequently applied to SAR data acquired during the deepwater horizon (DWH) oil spill in the Gulf of Mexico.**

***Index Terms*—Co-polarization difference (PD), marine oil spill, microwave scattering, non-Bragg scattering, oil concentration, synthetic aperture radar (SAR), uninhabited aerial vehicle synthetic aperture radar (UAVSAR).**

I. Introduction

MARINE oil spills refer to the release of oil and related substances into the marine environment, typically resulting from accidental discharges or improper human operations. These events commonly occur during oil extraction, transportation, storage, and utilization processes, including tanker accidents, offshore drilling platform leaks, pipeline ruptures, and mishandling during loading or unloading at ports. Preventing such pollution incidents remains challenging, particularly under adverse weather conditions, mechanical failures, or human errors. As a result, effective monitoring of oil spills on the ocean surface has become a critical area of research [1], [2], [3].

To mitigate the consequences of oil spills, remote sensing plays a key role in large-scale monitoring. Synthetic aperture radar (SAR), an active microwave remote sensing system, offers all-weather, day-and-night imaging capabilities, making it one of the most effective tools for marine oil spill detection [4], [5], [6], [7]. At microwave frequencies and moderate incidence angles, Bragg scattering dominates the radar backscattering from the sea surface [8], [9]. In this mechanism, short ocean surface waves with wavelengths comparable to that of the radar resonate with the incident electromagnetic waves. Oil spills reduce radar backscatter through two primary mechanisms [2], [10]: 1) suppression of capillary and short gravity-capillary waves, which are the main contributors to Bragg scattering; and 2) reduction of the effective dielectric constant of the sea surface when the oil film is sufficiently thick or emulsified due to the lower dielectric constant of oil relative to seawater. Consequently, oil-contaminated regions in SAR imagery typically appear darker than clean sea surfaces. Over the past few decades, extensive research on SAR-based oil spill detection and monitoring has been conducted. Although considerable progress has been made, significant challenges remain, particularly in achieving reliable quantitative assessments of oil spill characteristics from SAR data due to the complex nature of microwave scattering over oil-contaminated sea surfaces [2], [11].

Research on the quantitative inversion of marine oil spills has attracted significant attention, with various studies exploring different approaches. Minchew [12] proposed using the mixing index (Mdex) to assess the degree of oil–water mixing and the physical properties of oil. A lookup table and inversion algorithm were applied to calculate variations in surface roughness (M_w) and reflection coefficients (M_α), with the Mdex formula quantifying their relative contributions. Collins et al. [13] evaluated oil slick characteristics using Mdex and compared pseudo-quad-polarimetric SAR data with true

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Honglei Zheng and Yunhua Wang are with the Faculty of Information Science and Engineering, Ocean University of China, Qingdao 266100, China (e-mail: zhenghonglei@ouc.edu.cn; yunhuawang@ouc.edu.cn).

Peng Ren is with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: pengren@upc.edu.cn).

Weimin Huang is with the Faculty of Engineering and Applied Science, Memorial University of Newfoundland, St. John's, NL A1B 3X5, Canada (e-mail: weimin@mun.ca).

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quad-polarimetric SAR data to assess the potential of compact polarimetric (CP) SAR for studying oil–water mixing properties. Angelliaume et al. [10] analyzed dual-polarized echoes, specifically horizontal–horizontal (HH, i.e., horizontally polarized transmission and reception) and vertical–vertical (VV, i.e., vertically polarized transmission and reception), using the universal weighted curvature approximation (U-WCA) model to retrieve the dielectric constants of oil–water mixtures based on the co-polarization ratio (PR). Boisot et al. [14] developed an algorithm to estimate the volumetric fraction of oil–water mixtures from L-band SAR data. It uses the Elfouhaily spectrum to simulate wave conditions and applies the Bruggeman formula to compute the effective dielectric constant of oil–water mixtures. The U-WCA model simulates the PR, and solving it with the Bruggeman formula yields per-pixel estimates of oil–water volume fraction and dielectric constant. Li et al. [15] proposed a method to retrieve the oil–water mixing ratio from CP SAR imagery. That approach constructs a lookup table relating the oil–water mixing ratio to the ratio of diagonal elements (C11/C22) of the CP SAR covariance matrix. Based on the given incidence angle and observed data, the method retrieves the mixing ratio. The theoretical foundation lies in the assumption that the HH/VV ratio in CP SAR is sensitive to changes in dielectric constants while remaining independent of surface roughness. Quigley et al. [16], [17] proposed a polarimetric two-scale model (TSM) to retrieve the dielectric properties of oil slicks using Radarsat-2 SAR data. This model assumes that the ocean surface consists of large- and small-scale roughness, where small-scale roughness follows a zero-mean stochastic process, and large-scale roughness is characterized by tilted plane distributions. The polarimetric ratio eliminates small-scale roughness effects, making the model primarily dependent on dielectric constants and large-scale roughness. A lookup table was used to relate the polarimetric ratio to dielectric constants, which were then converted to oil–water volumetric fractions. Meng et al. [18] introduced a method combining an electromagnetic scattering model and artificial neural networks (ANNs) to retrieve oil slick thickness and oil–water volumetric fractions from SAR imagery. The advanced integral equation model (AIEM) was used to simulate microwave scattering characteristics of oil-covered and oil-free ocean surfaces, accounting for the damping effect of small-scale roughness and the influence of the complex dielectric constant of oil–water mixtures on radar backscatter. The model reflects the nonlinear relationship between oil slick thickness and volumetric fractions using the damping ratio (DR), with ANN applied to predict oil slick thickness and oil–water volumetric fractions.

Most existing oil concentration retrieval methods rely on the co-PR (HH/VV) derived from SAR data. Oil spills dampen small-scale waves at Bragg resonance wavelengths and reduce the effective dielectric constant of the sea surface, resulting in attenuation of the radar backscatter signal. The SAR co-PR is often used to decouple these two effects, as it is sensitive to changes in the dielectric constant while largely independent of surface roughness. A fundamental assumption in these studies is that sea surface scattering is dominated by the Bragg scattering mechanism. In reality, however, sea surface scattering comprises both Bragg and non-Bragg components. The latter is typically associated with breaking waves and is modulated by the presence of surface films [9], [19], [20], [21]. Non-Bragg scattering has a minor effect on VV-polarized returns but significantly influences HH-polarized returns, particularly at L-band frequencies. Due to the complex scattering processes involved in oil-covered sea surfaces, understanding the role of non-Bragg scattering in relation to oil properties—such as concentration—remains a significant challenge [2], [21].

In our previous work Zheng et al. [11], we proposed an algorithm to improve oil concentration retrieval by removing contributions from non-Bragg scattering. That study assumed that the relative contribution of non-Bragg scattering over oil-covered surfaces was the same as that over clean sea surfaces. However, this assumption may introduce uncertainty and potential errors. Currently, no theoretical model exists to describe the relationship between oil concentration and non-Bragg scattering, which introduces uncertainty in PR-based inversion methods operating under the Bragg scattering framework. To overcome this, we propose a retrieval method using co-polarization difference (PD), which is unaffected by non-Bragg scattering. This approach effectively removes the influence of non-Bragg scattering from the retrieval process. The proposed algorithm is validated using both simulated SAR data across varying oil concentrations and uninhabited aerial vehicle synthetic aperture radar (UAVSAR) data from the deepwater horizon (DWH) oil spill in the Gulf of Mexico.

This article is organized as follows. Section II introduces the theoretical background, including microwave scattering mechanisms over the sea surface, the damping effects of oil on short-scale ocean waves, and the influence of oil spills on the effective dielectric constant. Section III describes the datasets employed in this study. Section IV presents the data analysis and details the proposed PD-based retrieval algorithm. Section V provides validation results of the PD-based method along with discussions. Finally, Section VI concludes the article with a summary of key findings and implications.

II. Microwave Scattering Theory for Oil-Covered Sea Surfaces

A. Bragg Scattering From Sea Surface

Microwave scattering modeling of the sea surface has a long history, during which several classical approaches have been developed to characterize the interaction between electromagnetic waves and ocean surface roughness. Among these, the small-slope approximation (SSA) [22], the TSM [23], and the AIEM [24] are widely employed to simulate microwave backscattering from the sea surface.

Within the framework of the TSM, the Bragg normalized radar cross section (NRCS) σ_{pp}^{B} at moderate incidence angles can be approximated using a first-order Taylor expansion with respect to the slope of long waves [8]

$$\sigma_{\mathrm{pp}}^{B} = \frac{\pi}{\tan^4\theta_i}\left|G_{\mathrm{pp}}\right|^2\left(1+g_{\mathrm{pp}}s_i^2\right)^2 B\left(k_B,\varphi\right) \tag{1}$$

where pp represents either horizontal–horizontal (HH) or vertical–vertical (VV) polarization. θ_i represents the incidence

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angle, and G_{pp} and g_{pp} can be written as

$$\begin{cases} g_{HH} = \dfrac{\tan^4\theta_i}{2|G_{HH}|^2}\dfrac{\partial^2}{\partial\theta_i^2}\left(\dfrac{|G_{HH}|^2}{\tan^4\theta_i}\right) + \dfrac{2}{\sin^2\theta_i}\left|\dfrac{G_{VV}}{G_{HH}}\right|\dfrac{S_n^2}{S_i^2} \\ g_{VV} = \dfrac{\tan^4\theta_i}{2|G_{VV}|^2}\dfrac{\partial^2}{\partial\theta_i^2}\left(\dfrac{|G_{VV}|^2}{\tan^4\theta_i}\right) \end{cases} \tag{2}$$

$$\begin{cases} G_{HH} = \dfrac{\varepsilon_r - 1}{\left[\cos\theta_i + \sqrt{\varepsilon_r - \sin^2\theta_i}\right]^2} \\ G_{VV} = \dfrac{(\varepsilon_r - 1)\times\left[\varepsilon_r\left(1+\sin^2\theta_i\right) - \sin^2\theta_i\right]}{\left[\varepsilon_r\cos\theta_i + \sqrt{\varepsilon_r - \sin^2\theta_i}\right]^2} \end{cases} \tag{3}$$

where ε_r represents the relative dielectric constant of seawater. s_i^2 and s_n^2 are the mean square slope (MSS) of tilting waves in and out of the direction of the incidence plane, respectively. $k_B = 2k_i\sin\theta_i$ is the Bragg wavenumber and k_i is the radar wavenumber. The directional curvature spectrum is expressed as $B(k_B,\varphi) = B_0(k_B)f(k_B,\varphi)$, where $B_0(k)$ is the omnidirectional part, $f(k_B,\varphi)$ is the angular spreading function, and φ represents the angle between the wind direction and the radar look direction. The relationship between the curvature spectrum and the sea wave elevation spectrum can be expressed as $B_0(k_B) = k_B^3 S_0(k_B)$.

SAR can be regarded as a spectral filter for ocean waves, where only those waves with wavenumbers resonant with the radar wavelength contribute significantly to the backscattered signal. By combining (1) and (2), the curvature spectrum of Bragg-resonant waves can be derived from SAR observations

$$B(k_B,\varphi) = \frac{(\sigma_{VV} - \sigma_{HH})\tan^4\theta_i}{\pi\left[|G_{VV}|^2\left(1+g_{VV}s_i^2\right) - |G_{HH}|^2\left(1+g_{HH}s_i^2\right)\right]}. \tag{4}$$

Over the past decades, various sea surface wave spectral models have been proposed, including the Elfouhaily spectrum [25], the Hwang spectrum [26], and the Apel spectrum [27]. These models are widely used for simulating microwave scattering from the sea surface [28], [29], [30]. These spectral predictions encompass all short-scale roughness contributions, including those associated with breaking wave areas [8]. In contrast, the Bragg spectrum estimated from SAR images using (4) captures only Bragg-resonant waves.

B. Suppression of Ocean Waves Induced by Oil Spills

Oil film on sea surface damps capillary waves and small-scale gravity waves; this phenomenon has been extensively studied [31], [32], [33], [34], [35]. When considering only the damping effect for small-scale waves, the sea spectrum damping coefficient (DC)—defined as the ratio of the wave spectrum of a clean sea surface to that of an oil-covered sea surface—can be modeled using Marangoni damping theory [31], [36], [37], [38]

$$y(k) = \frac{1 + X(\cos\delta - \sin\delta) + XY - Y\sin\delta}{1 + 2X(\cos\delta - \sin\delta) + 2X^2} \tag{5}$$

$$X = \frac{|E|k^2}{\sqrt{2\omega^2\eta\rho}}, \quad Y = \frac{|E|k}{4\eta\omega} \tag{6}$$

where k is the spatial wavenumber of sea waves, $|E|$ and δ represent the magnitude and phase, respectively, of the complex dilational modulus of the surface oil film. In this work, $|E| = 0.01\,\mathrm{N\cdot m^{-1}}$, $\delta = 175°$ are adopted [39]. ω is the angular frequency of the sea surface waves. η and ρ are the dynamic viscosity and density of seawater, respectively.

C. Reduction in Sea Surface Dielectric Constant Due to Oil Spills

When the oil film is sufficiently thick (at least one-tenth of the radar wave penetration depth—for L-band radar, at least 4 mm [40]) or emulsified with seawater, the effective dielectric constant of the sea surface decreases, as oil has a lower dielectric constant than seawater. In the microwave range, the relative dielectric constant of crude oil varies within a narrow range, with an average value of $\varepsilon_{oil} = 2.3 - 0.02j$ [41], [42]. This value is significantly lower than that of seawater in the microwave frequency band. For example, at the L-band, the relative dielectric constant of seawater is approximately $\varepsilon_{sw} = 73.0 - 65.1j$ at a temperature of 15 °C and with a salinity of 35 PSU [10]. The Bruggeman formula can be employed to simulate the effective dielectric constant ε_{em} of the oil–water mixture [10], [42]

$$\varepsilon_{em} = \frac{1}{4}\left\{\varepsilon_{sw} - (1-3M)(\varepsilon_{oil} - \varepsilon_{sw}) + \sqrt{[\varepsilon_{sw} - (1-3M)(\varepsilon_{oil} - \varepsilon_{sw})]^2 + 8\varepsilon_{sw}\varepsilon_{oil}}\right\} \tag{7}$$

where M denotes the oil concentration (oil volume fraction of the oil–water mixture).

D. Effect of Non-Bragg Scattering

At present, most oil spill concentration retrieval methods are based on co-PR (HH/VV). These methods compute the theoretical Bragg PR corresponding to different oil concentrations, establish a mapping between oil concentration and PR, and retrieve per-pixel oil concentrations by comparing observed SAR PR values with theoretical Bragg PR values. The primary advantage of this approach is that the Bragg PR is independent of the sea surface wave spectrum, enabling the decoupling of wave damping effects and changes in the dielectric constant. However, a major limitation lies in the discrepancy between the theoretical Bragg PR and the actual SAR-derived PR. As shown in Fig. 1, which compares the Bragg PR with measured SAR PR along the red transect in Fig. 2, a substantial mismatch is observed. This inconsistency can introduce significant errors in oil concentration retrieval.

The discrepancy between the Bragg PR and SAR PR can be attributed to the influence of non-Bragg scattering. To incorporate the effects of non-Bragg scattering, NRCS σ_{pp} of sea surface at moderate incidence angles can be expressed as [8]

$$\sigma_{pp} = \sigma_{pp}^{B} + \sigma^{nB} \tag{8}$$

where σ_{pp}^{B} denotes the NRCS of Bragg scattering, and σ^{nB} denotes the NRCS of non-Bragg scattering. The non-Bragg scattering component is typically associated with breaking waves and is independent of polarization [9], [19], [20].

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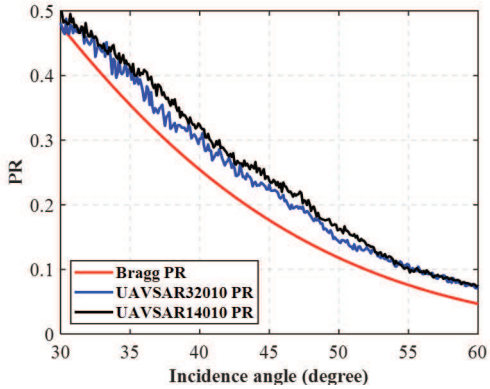


Fig. 1. PR calculated based on (2) and UAVSAR data.

The co-PD is defined as

$$\mathrm{PD} = \sigma_{\mathrm{VV}} - \sigma_{\mathrm{HH}} = \sigma^{B}_{\mathrm{VV}} - \sigma^{B}_{\mathrm{HH}}. \tag{9}$$

The PD eliminates the non-Bragg component of the backscattered signal, primarily capturing contributions from short waves near the Bragg wavenumber. These waves are directly influenced by the presence of an oil film, making PD a valuable parameter for oil slick detection at sea [21], [43]. Building on this, we develop a new oil concentration retrieval method based on PD, which effectively reduces errors caused by non-Bragg scattering.

III. Dataset

The dataset employed in this study was acquired by the UAVSAR, a fully polarimetric L-band SAR system operating at a center frequency of 1.2575 GHz. The data used here are identical to those utilized in our previous study [11]. For completeness, a brief overview is provided below.

A. DWH Oil Spill Dataset

Fig. 2 displays VV-polarized NRCS images (in dB) derived from UAVSAR data collected during the 2010 DWH oil spill. Specifically, Fig. 2(b), referred to as UAVSAR32010, corresponds to the area outlined by the blue rectangle in Fig. 2(a), while Fig. 2(c), referred to as UAVSAR14010, corresponds to the area outlined by the red rectangle in Fig. 2(a). The dataset was obtained under wind speeds of approximately 2.5–5 m/s, with wind directions ranging from 115° to 126° [41]. Although this dataset was collected over a decade ago, the UAVSAR sensor continues to be actively used for remote sensing of the Earth's surface, ensuring the ongoing research value of this data. The signal-to-noise ratio (SNR) is a crucial factor in studying scattering properties using SAR data. Minchew et al. [41] analyzed the effects of noise on radar backscatter. In this study, only co-polarized data (HH and VV) are used, and the impact of noise is negligible when the incidence angle is smaller than approximately 62°.

B. NORSE2015 Experiment Dataset

Fig. 3 shows VV-polarized NRCS images (in dB) from the NORSE2015 experiment. Fig. 3(a) and (b) corresponds to UAVSAR0626 and UAVSAR0639, respectively. Four distinct

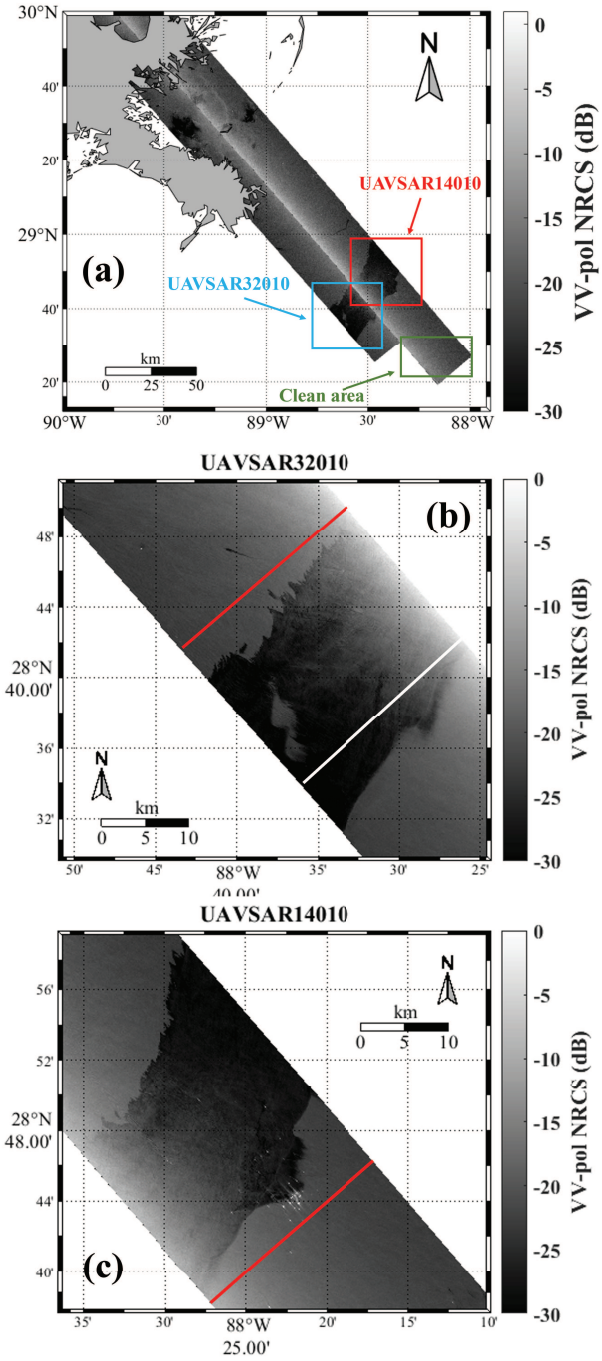


Fig. 2. (a) Overview of the DWH oil spill area. The red and blue rectangles denote the spatial coverage of the UAVSAR14010 and UAVSAR32010 scenes, respectively. The green rectangle marks a clean sea reference area. (b) UAVSAR32010 and (c) UAVSAR14010.

dark patches are visible in both images, corresponding to from top to bottom: emulsified oil with 80% crude oil mixed with seawater (E80, $M = 0.8$), 60% crude oil (E60, $M = 0.6$), 40% crude oil (E40, $M = 0.4$), and plant oil. The dataset was collected under wind speeds of approximately 10–12 m/s. For further details on the NORSE2015 experiment, refer to [44], [45], and [46].

To suppress speckle noise, a 7×7 boxcar (moving average) filter was applied to both Dataset A and Dataset B during preprocessing.

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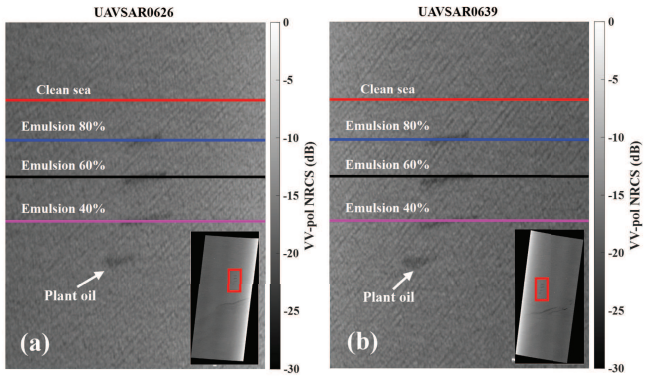


Fig. 3. NORSE2015 experiment dataset used in this work (a) UAVSAR0626 and (b) UAVSAR0639. An overview image is displayed in the bottom-right corner of each panel. These overviews indicate the spatial locations of the extracted subareas used in subsequent analysis.

C. Simulated UAVSAR Data of Oil Spills With Varying Concentrations

Considering the lack of reliable in situ measurements to validate the performance of the proposed inversion algorithm, we simulate the NRCS of oil-covered sea surfaces at various concentrations based on real UAVSAR observations. These simulations serve as a reference for evaluating the proposed algorithm in Section V-A. The oil spill data at different concentrations are simulated using the following procedure. First, a uniform clean sea surface region within the green rectangle in Fig. 2(a) is selected. The curvature spectrum for each pixel is then computed using (4). By dividing this curvature spectrum by the DC obtained from (5), the curvature spectrum corresponding to oil-covered sea surfaces is derived. Subsequently, by incorporating the dielectric constant of oil spills at different concentrations—calculated using (7)–into (1), the Bragg NRCS of oil-covered sea surfaces is computed. Finally, the non-Bragg NRCS component is added to each pixel. However, the NRCS associated with non-Bragg scattering from thick oil-covered sea surfaces–where both surface wave damping and the reduction in the dielectric constant contribute to radar backscatter attenuation—remains uncertain. To examine the impact of non-Bragg scattering on oil concentration retrieval, we simulate the non-Bragg NRCS using two approaches.

1) The relative contribution of non-Bragg scattering ($\sigma^{\mathrm{nb}}/\sigma_{\mathrm{pp}}$) of the oil-covered sea surface is identical to that of the clean sea surface.
2) The non-Bragg NRCS (σ^{nb}) of the oil-covered sea surface is identical to that of the clean sea surface.

The non-Bragg NRCS of the clean sea surface is calculated using (8). Since thermal noise is polarization-independent and the non-Bragg component is also polarization-independent, the contribution of thermal noise is inherently included in the non-Bragg NRCS during the decomposition process, although its magnitude is very small. Finally, the total HH- and VV-polarized NRCS values are obtained by summing the Bragg and non-Bragg components for each pixel.

Fig. 4 presents the original clean sea surface data from the region outlined in green in Fig. 2, along with VV-polarized

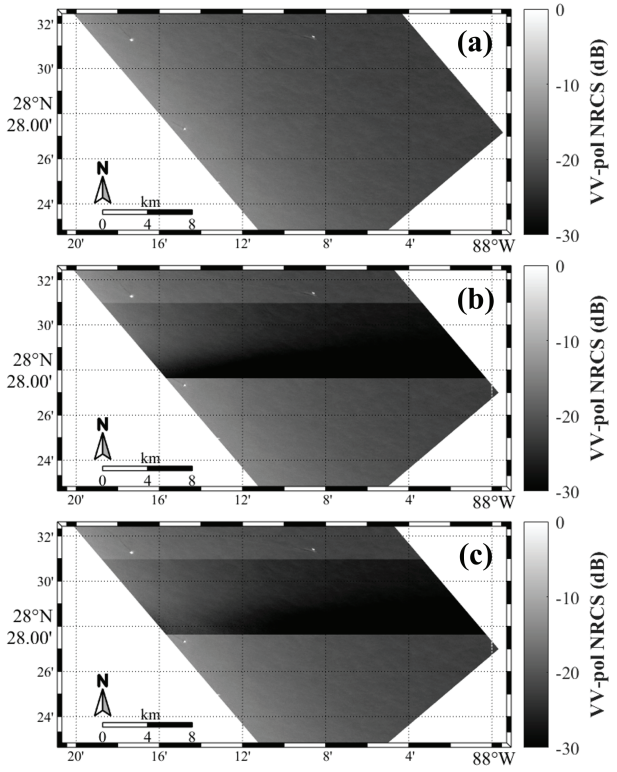


Fig. 4. Simulated UAVSAR data of oil spills with varying concentrations. (a) Original SAR data. (b) Simulated oil data with assumption (1). (c) Simulated oil data with assumption (2).

NRCS maps of the simulated oil spill under assumptions (1) and (2). The dark areas in Fig. 4(b) and (c) represent simulated oil-covered regions. In these simulations, the oil concentration is uniform along the horizontal direction and increases linearly from 0 to 1 along the vertical axis, with a step size of 0.001. The simulated results shown in Fig. 4 serve as a synthetic reference for evaluating the performance of the proposed retrieval method. Detailed analysis and interpretation of these results are provided in Section V-A.

IV. PD-Based Method for Estimating Oil Concentrations

In general, oil released into the marine environment typically appears in two forms: thin oil films or water-in-oil emulsions [2], [10], [47]. In the first case, when the oil film thickness is smaller than the radar wave penetration depth, radar waves penetrate the film and scatter at the oil–water interface. Consequently, the effective dielectric constant of the sea surface remains same as that of seawater, and radar echo attenuation is primarily due to the suppression of small-scale waves. In the second case, involving water-in-oil emulsions, the oil spill not only dampens small-scale waves but also reduces the effective dielectric constant of the sea surface, further weakening the radar echo. SAR-based oil concentration retrieval is possible only in this scenario and only when the oil thickness exceeds one-tenth of the radar penetration depth [2], [40], which defines the conditions examined in this study. For water-in-oil emulsions, if the contribution of wave suppression to radar echo reduction can be quantified, the

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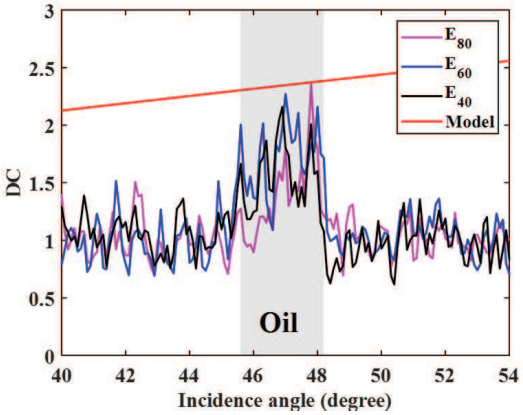


Fig. 5. DCs calculated from the Marangoni model and the NORSE2015 experimental dataset. In the legend, E80, E60, and E40 represent the DC values calculated along straight lines passing through oil spill regions with different concentrations in Fig. 3(a). The shaded areas correspond to the oil-covered regions.

remaining attenuation can be attributed to the decrease in the dielectric constant. Based on this principle, oil concentration can be retrieved. However, the extent to which oil spills of different concentrations dampen ocean waves remains uncertain. Studying the damping effect of thick oil spills with varying concentrations is particularly challenging due to the lack of available data and theoretical models.

A. Damping Effect Induced by Oil Spill With Different Oil Concentrations

The NORSE2015 experiment dataset, shown in Fig. 3, is one of the few available datasets for studying the microwave scattering characteristics of oil spills at different oil concentrations. Since the estimated emulsion thickness in the NORSE2015 experiment is in the range of 1.3–1.7 μm (significantly smaller than the penetration depth of L-band radar, which is 7–9 mm) [44], the attenuation of echoes in the oil spill area is solely due to the suppression of small-scale waves caused by the oil. In this case, DC is calculated as the ratio of the PDs between the clean sea surface and the oil-covered sea surface. Fig. 5 presents the DC values, where the shaded areas indicate regions in Fig. 3(a) that intersect with the oil spill. The curves corresponding to different oil spill concentrations appear rough due to the influence of speckle noise. The red line in Fig. 5 represents the simulated values obtained using the Marangoni DC theoretical model described in (5).

In Fig. 5, the differences among the DC values corresponding to different oil concentrations are not significant, with maximum values around 2. Additionally, the DC values for all three concentrations are slightly lower than those predicted by the Marangoni damping model. The NORSE2015 experiment dataset used in Fig. 5 was collected under sea conditions with a wind speed of 12 m/s. Under high sea states, the damping effect of oil spills on ocean waves weakens [41]. In other words, at low to moderate wind speeds, the DC may be higher than the values shown in Fig. 5. Due to the lack of measured data at different oil concentrations under low to moderate wind speeds, assessing the discrepancy between the theoretical values predicted by the Marangoni model and the actual values in such conditions remains challenging. The results presented in Fig. 5 suggest that the Marangoni model yields reasonably accurate predictions of the DC.

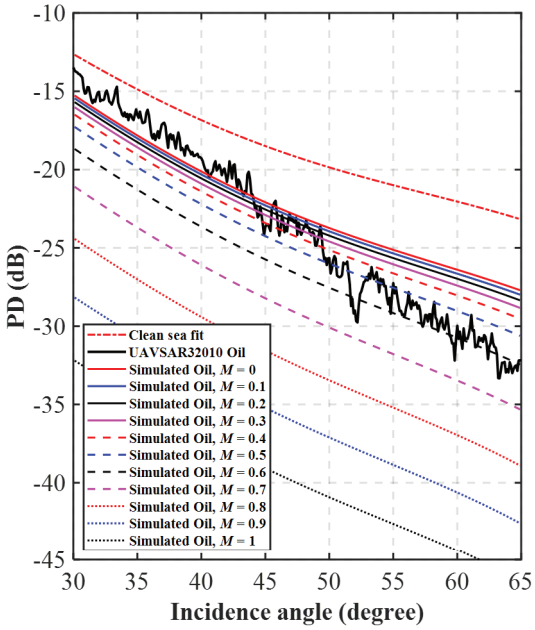


Fig. 6. PDs over different sea surfaces.

B. Effects of Small-Scale Wave Suppression and Dielectric Constant Reduction on PD Over Oil-Covered Sea Surfaces

In this section, we investigate the effects of damping effect and the reduction in the effective dielectric constant on PD through theoretical simulations. First, we extract the NRCS of the clean sea surface along the red line in Fig. 2(b). The Bragg curvature spectrum is then computed using (4), followed by incidence angle fitting to eliminate the influence of speckle noise. The fit curvature spectrum, along with the dielectric constants corresponding to different oil concentrations from (7), is then substituted into (1) to determine the Bragg NRCS for HH and VV polarizations. Finally, the PD for various oil concentrations is obtained by computing the difference between HH and VV polarizations.

Fig. 6 illustrates the PD over different sea surfaces. The PD of the clean sea surface (labeled as "clean sea fit") is extracted along the red line in Fig. 2(b) and fit to the incidence angle, resulting in a smooth curve. In contrast, the PD of the oil spill area (labeled as "UAVSAR32010 Oil") is extracted along the white line in Fig. 2(b) without incidence angle fitting. As a result, it is affected by speckle noise, making the curve appear rough. The simulated PD values of oil spills with various concentrations correspond to the curves labeled as "simulated oil, $M = 0 \sim 1$." The $M = 0$ curve represents a scenario where PD (as well as NRCS) reduction is solely due to the damping effect. When M takes nonzero values, PD reduction is influenced by both wave suppression and changes in the

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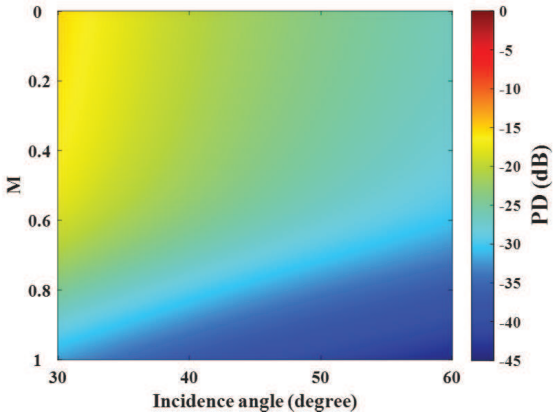


Fig. 7. PD lookup table generated for various oil concentrations and incidence angles, based on the DWH dataset.

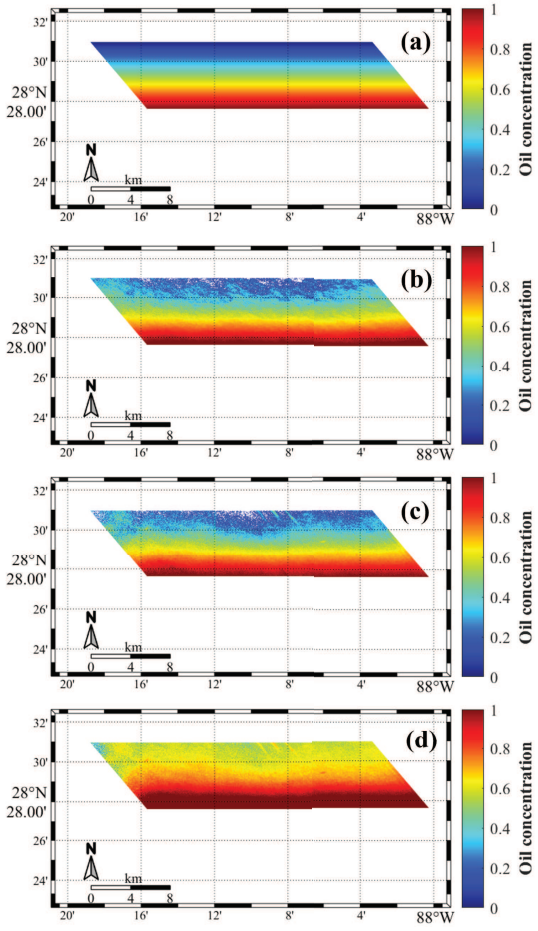


Fig. 8. (a) Preset oil concentration values in the simulated dataset. (b) Inversion results using the proposed PD-based method. (c) and (d) Inversion results for simulated cases (1) and (2) from Fig. 4(b) and (c), respectively, using the same method.

dielectric constant. As oil concentration increases, the effective dielectric constant of the oil-covered sea surface decreases, leading to further PD reduction. This reduction becomes more pronounced at higher oil concentrations, indicating that PD's sensitivity to oil concentration variations increases with oil concentration. This property of PD is similar to that of PR,

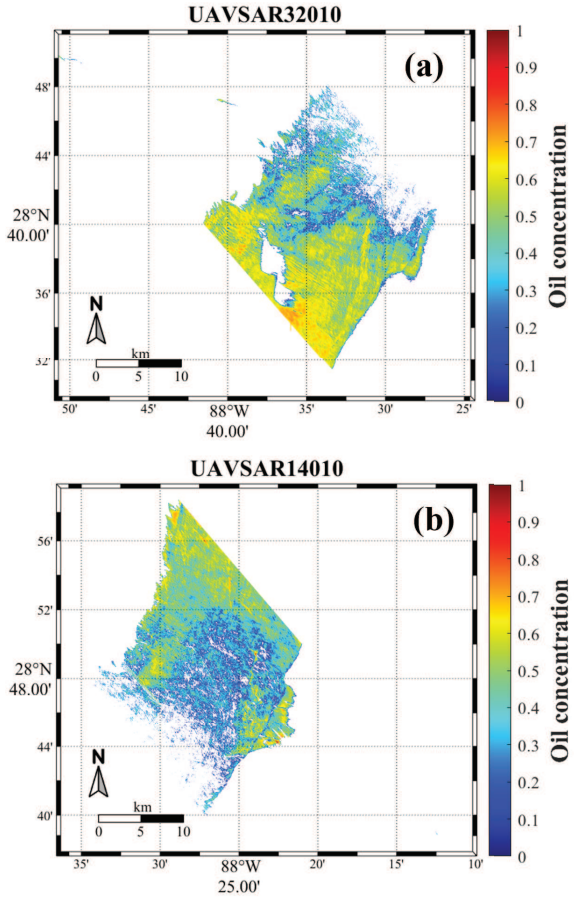


Fig. 9. Oil concentration retrieval results using the PD-based method. (a) UAVSAR32010. (b) UAVSAR14010.

as shown in Fig. 10 of [10]. This suggests that when the oil spill concentration is low, both the PD and the PR are sensitive to concentration variations. Consequently, retrieving low-concentration oil spills using PD and PR may be highly susceptible to speckle noise, potentially leading to inaccurate estimations.

In Fig. 6, when the incidence angle is below about 44°, the PD of the oil spill area (labeled as "UAVSAR32010 Oil") is higher than that of the $M = 0$ curve. This suggests that the oil film coverage within the resolution cell is incomplete (i.e., a low filling factor), resulting in a partial damping effect on short-scale waves. When oil is released into the marine environment, it initially spreads across the sea surface, forming a relatively uniform film. However, due to waves, currents, and turbulence, the oil film often fragments into multiple patches. The spatial distribution of oil can be quantified using the fractional filling factor, which represents the proportion of an SAR image resolution cell covered by the oil film [31]. This study focuses on thick oil, where both the damping effect and the reduction in the dielectric constant contribute to the NRCS decrease. This corresponds to the region in Fig. 6 where the incidence angle exceeds 44°. By comparing the PD of the oil-covered region (the curve labeled as "UAVSAR32010 Oil") with the PD curves under various oil concentrations, the oil spill concentration can be determined.

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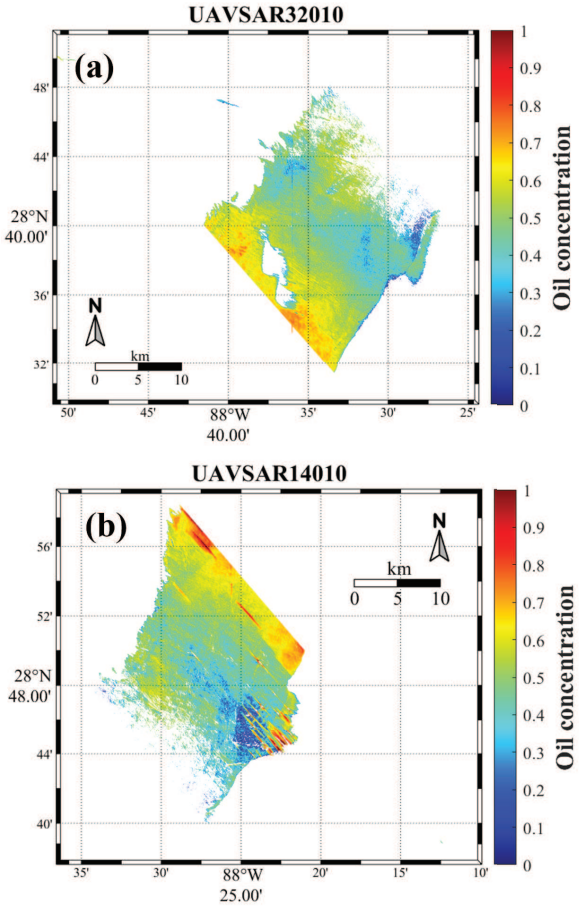


Fig. 10. Oil concentrations inverted using the PR-based method in [11]. (a) UAVSAR32010. (b) UAVSAR14010.

C. PD-Based Method

According to the discussions in Section IV-A, the Marangoni damping model can be used to approximate the DC for oil spills of different concentrations. By isolating the portion of radar echo attenuation caused by the damping effect, the remaining attenuation can be attributed to the reduction in the effective dielectric constant. Following this principle, we propose the following steps to estimate oil spill concentration.

1) Calculating the Bragg Curvature Spectrum of the Clean Sea Surface Under Different Incidence Angles: To ensure a wide range of incidence angles, a clean sea surface region is visually selected from the SAR image (such as the red line region in Fig. 2). The HH- and VV-polarized NRCS values of the clean sea surface are then extracted. Using (4), the Bragg curvature spectrum of the clean sea surface is estimated for different incidence angles. To mitigate the influence of speckle noise, the curvature spectrum is fit to the incidence angle, resulting in a smooth curve across different incidence angles.

2) Calculating the Bragg Curvature Spectrum of the Oil-Covered Sea Surface: The Bragg curvature spectrum of the oil-covered sea surface is obtained by dividing the Bragg wave spectrum of the clean sea surface by DC, which is calculated using the Marangoni damping model described in (5).

3) Constructing a PD Lookup Table for Different Oil Concentrations and Incidence Angles: The Bragg curvature spectrum of the oil-covered sea surface, along with the dielectric constants for different oil concentrations [see (7)], is substituted into (2) to obtain the Bragg NRCS for HH and VV polarizations. The difference between these coefficients provides the PD for various oil concentrations and incidence angles. The results are compiled into a lookup table. Fig. 7 presents a PD lookup table generated for various oil concentrations and incidence angles, based on the DWH oil spill dataset. The oil concentration M is sampled with a step size of 0.001, and the incidence angle sample interval is 0.1 radians.

4) Pixel-by-Pixel Estimation of Oil Spill Concentration: The PD and incidence angle for each pixel within the oil spill region are directly extracted from the SAR data. By comparing the SAR-derived PD of each pixel with the lookup table, the corresponding oil concentration can be determined.

V. Validation and Discussion

A. Validation of the PD-Based Method Using Simulated Data

We applied the PD-based method to the simulated SAR data shown in Fig. 4 and compared the inversion results with those obtained using the PR-based method from [11]. The preset M values in the simulated SAR data are shown in Fig. 8(a), while Fig. 8(b) presents the inversion results using the PD-based method. In Fig. 4, we simulated the non-Bragg NRCS of oil-covered sea surfaces using two different approaches. Since the PD-based method completely eliminates the influence of non-Bragg scattering, the inversion results for both simulated datasets are identical. As shown in Fig. 8(b), the PD-based inversion method accurately estimates high-concentration oil spills. However, at lower concentrations, discrepancies arise between the inversion results and the true values. This is because, at lower oil spill concentrations, PD is highly sensitive to concentration variations, which aligns with the results shown in Fig. 6.

Fig. 8(c) and (d) presents the inversion results obtained using the PR-based method. Specifically, Fig. 8(c) corresponds to the inversion results for Fig. 4(b), while Fig. 8(d) displays the results for Fig. 4(c). Similar to Fig. 8(b) and (c) shows that the inversion results align well with the preset values at higher concentrations but are significantly affected by speckle noise at lower concentrations. In contrast, Fig. 8(d) presents the PR-based inversion results for the simulated dataset under assumption (2), where the non-Bragg NRCS of the oil-covered region is assumed to be the same as that of the clean sea surface. In this case, the inversion results tend to be overestimated. At present, the exact contribution of non-Bragg scattering to oil-covered sea surfaces remains uncertain, which is why we employed two different assumptions to simulate oil spill data. Fig. 8 reveal that when the non-Bragg scattering characteristics of oil spills are not well understood, the PR-based method does not yield reliable results. In contrast, the PD-based method effectively eliminates the influence of non-Bragg scattering, producing more accurate inversion results. This highlights the advantage of using the PD over the PR for oil spill concentration estimation when the effects of non-Bragg scattering are uncertain.

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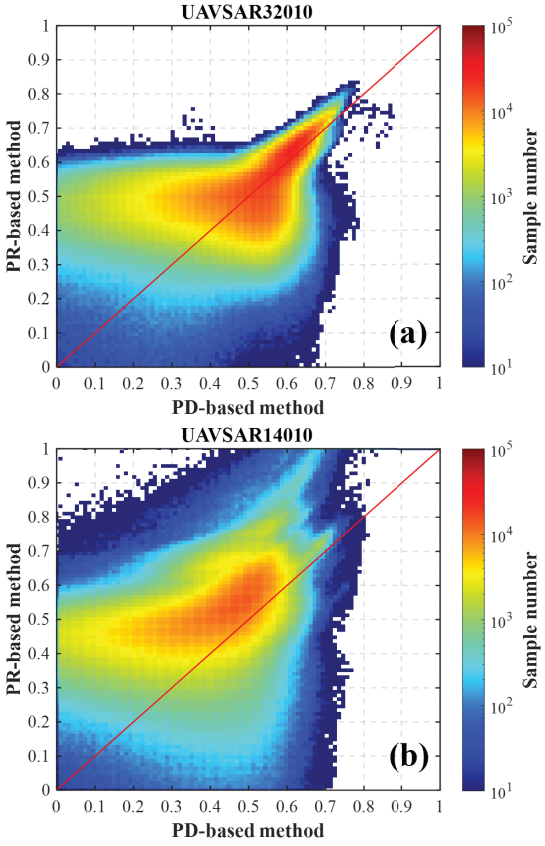


Fig. 11. Scatter plots of the retrieval results of the PD-based method and the PR-based method, with the PR-based results shown in Fig. 2. (a) UAVSAR32010. (b) UAVSAR14010.

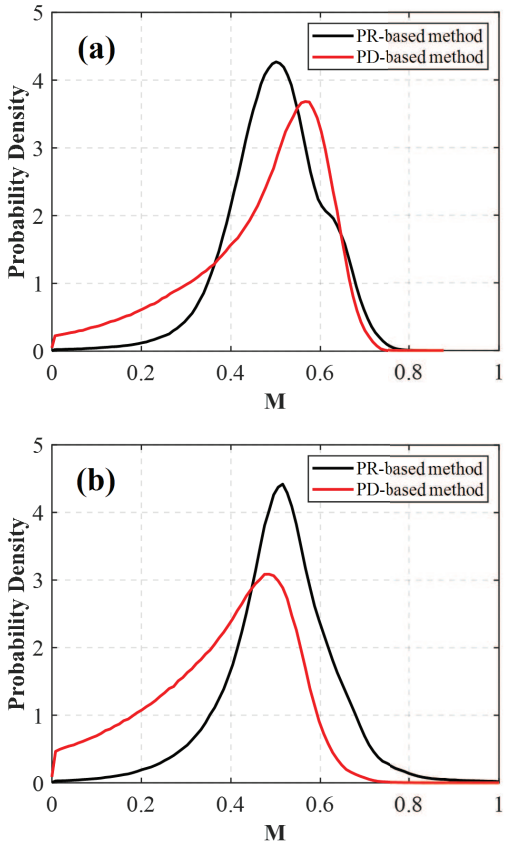


Fig. 12. Comparison of the probability density distributions between the retrieval results of the PD-based method and the PR-based method. (a) UAVSAR32010. (b) UAVSAR14010.

B. Application of the PD-Based Method to the DWH Oil Spill Dataset

Figs. 9 and 10 presents the inversion results obtained using the PD-based and PR-based methods for the DWH oil spill dataset. While the overall distribution of the inverted M values is generally similar, differences exist in finer details. Fig. 11 displays a scatter plot comparing the results from both methods. In Fig. 11, the results from both methods show a strong positive correlation, with most data points clustering around the red diagonal line, indicating overall consistency between the approaches. Fig. 12 presents the probability density distributions of M values obtained using the PD-based and PR-based methods. In both cases, the PR-based method produces a sharper and more concentrated distribution, with a pronounced peak near $M \approx 0.55$, suggesting that this method provides more consistent estimates centered around this value. In contrast, the PD-based method yields a broader distribution with a lower peak, potentially capturing more details of oil concentration variations.

C. Discussion

At present, most oil concentration inversion algorithms rely on the PR within the framework of Bragg scattering theory, where the influence of non-Bragg scattering is typically neglected. In Zheng et al. [11], we proposed a retrieval method based on the PR to decouple the effects of wave suppression and changes in the dielectric constant. While that approach is consistent with mainstream methods, it introduces a novel step to eliminate the influence of non-Bragg scattering over oil-covered sea surfaces. However, this additional step may introduce uncertainty and potential errors, as the relative contribution of non-Bragg scattering from oil spills with varying concentrations remains poorly characterized. Unlike the PR-based method, the PD-based method completely eliminates the impact of non-Bragg scattering. In this study, the proposed method is developed and validated using L-band SAR data, where the contribution of non-Bragg scattering is relatively smaller compared to higher frequency microwave bands such as C, X, and Ku [48]. Since non-Bragg scattering becomes more significant at higher frequencies, the advantages of the PD-based inversion algorithm are expected to be more pronounced when applied to C-, X-, or Ku-band systems.

However, the PD-based method introduces a new source of error related to DC. This study focuses on scenarios where the oil spill thickness exceeds one-tenth of the radar wave penetration depth, as only under these conditions does the change in the sea surface dielectric constant caused by the oil spill have a significant impact on microwave scattering. In this context, the relationship between DC, oil thickness, and concentration remains unknown. In this work, we use DC values simulated with the Marangoni DC model, which may introduce potential errors. In fact, both the PR-based method from [11] and the PD-based method proposed in

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this study have inherent limitations. Given the lack of field measurements, it remains uncertain which method provides more accurate results.

Accurately retrieving the oil concentration of marine oil spills is inherently challenging due to the complexity of microwave scattering from oil-covered sea surfaces. Even under clean sea surface conditions, factors such as steep waves, foam, breaking waves, and other dynamic processes make microwave scattering highly intricate, rendering the classic Bragg scattering mechanism insufficient to fully describe the scattering process. When oil covers the sea surface, it damps small-scale waves, alters the dielectric constant, reduces wind energy input, and increases viscous dissipation, among other effects. Moreover, these factors are influenced by oil properties, weathering time, and sea conditions, and they are interrelated and coupled, further complicating the retrieval process. Additionally, mineral oil films can form multilayered structures with varying thicknesses within the same oil patch [2]. The uneven distribution of oil thickness and coverage further complicates the inversion of oil spill concentrations. Current studies on oil spill concentration retrieval often rely on highly simplified scattering models to describe microwave interactions with oil-covered sea surfaces. These simplifications overlook several factors, including uneven oil thickness, nonuniform oil mixing, and non-Bragg scattering, which may introduce uncertainties and potential errors. Against this background, despite the limitations of the PD-based method proposed in this study, this work represents a step forward in quantitative oil spill concentration retrieval, as it explicitly accounts for factors such as non-Bragg scattering. Additionally, the proposed approach is modular, allowing for future refinements and improvements.

In this study, the retrieval results of the proposed method have not been validated against in situ measurements. The NORSE2015 experimental dataset is not suitable for validation purposes, as the oil slicks in that experiment were extremely thin and did not significantly alter the effective dielectric constant of the sea surface. Moreover, to the best of our knowledge, there are currently no available field datasets that can be directly used for validating this method. In fact, evaluating oil spill retrieval algorithms remains a challenging task. Direct validation using field measurements is rarely performed due to the inherent difficulties associated with measuring oil concentration in situ. The constant motion of ocean waves makes it nearly impossible to obtain data perfectly synchronized with SAR acquisitions. When field sampling data are used for validation, determining the appropriate sampling depth presents an additional problem, as it must align with the radar's electromagnetic wave penetration depth while accounting for continuous wave movement. Although controlled microwave scattering experiments, conducted either indoors or in coastal environments, may offer an intuitive pathway to validate oil spill concentration retrieval algorithms, their practical implementation remains highly challenging. In indoor tank experiments, generating realistic surface wave conditions typically requires the use of mechanical wave generators. Under such dynamic conditions, the oil slick undergoes continuous redistribution, and accurately measuring its spatially varying concentration and thickness in real time becomes extremely difficult. Synchronizing these measurements with microwave scattering observations adds further complexity. Alternatively, experiments conducted over calm water surfaces fail to replicate the dominant scattering mechanisms such as Bragg or composite scattering, which are responsible for SAR backscatter from ocean surfaces. Instead, they predominantly exhibit quasi-specular reflection, which limits the representativeness of such results for SAR-based inversion methods. To date, no existing study has reported successful validation of SAR-derived oil concentration using controlled scattering experiments with verified ground truth. Therefore, while such experiments may contribute to understanding underlying scattering physics, they are not yet suitable for direct validation of concentration retrieval performance. Without addressing these challenges, current approaches to oil concentration estimation can only be developed based on existing microwave scattering theoretical models.

VI. Conclusion

Retrieving marine oil spill concentrations remains a critical yet challenging research problem. To address this issue, a PD-based oil spill concentration retrieval method is proposed and evaluated using both simulated SAR data and DWH oil spill data. Comparative results with the traditional PR-based method demonstrate the feasibility and effectiveness of the PD-based approach. The key conclusions of this study are as follows.

1) The DC corresponding to different oil concentrations is estimated using the Marangoni damping model. However, as the model lacks a rigorous quantitative relationship between DC and oil concentration, this approximation introduces uncertainty in the retrieval results, representing the primary limitation of the PD-based method.

2) Both PD-based and PR-based methods have inherent limitations. Due to the absence of in situ measurement data, it remains unclear which approach yields higher retrieval accuracy. Nevertheless, the PD-based method offers a novel perspective on SAR-based oil spill concentration retrieval.

3) The retrieval accuracy of both methods improves with increasing oil concentration. This is because, at higher concentrations, PD and PR are less sensitive to background noise, leading to more robust and stable performance.

The PD-based method features a modular structure, allowing flexibility for future refinements. If a more precise relationship between DC and oil concentration can be established, the performance of the method is expected to improve. Therefore, future research should focus on acquiring reliable field measurements of oil spills at varying concentrations to support the development of more accurate retrieval algorithms.

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Honglei Zheng (Senior Member, IEEE) received the Ph.D. degree in marine detection technology from the Ocean University of China, Qingdao, China, in 2019.

From September 2017 to September 2018, he was a Visiting Ph.D. Student with Lab-STICC, ENSTA Bretagne, Brest, France. From September 2019 to May 2022, he was a Post-Doctoral Fellow with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao. Since May 2022, he has been an Associate Professor with the Faculty of Information Science and Engineering, Ocean University of China. Since December 2024, he has also been a Visiting Scholar with the Faculty of Engineering and Applied Science, Memorial University of Newfoundland, St. John's, NL, Canada.

His research interests include electromagnetic scattering modeling of rough sea surfaces and quantitative retrieval of marine oil spill parameters using microwave remote sensing.



Yunhua Wang was born in Shandong, China, in 1978. He received the M.S. and Ph.D. degrees in radio science from Xidian University, Xi'an, China, in 2005 and 2006, respectively.

He is currently a Professor with the Ocean Remote Sensing Institute, Ocean University of China, Qingdao, China. His research interests include asymptotic and numerical simulations of electromagnetic wave scattering from random sea surfaces and microwave ocean remote sensing.



Peng Ren (Senior Member, IEEE) received the B.S. and M.S. degrees in electronic engineering from Harbin Institute of Technology, Harbin, China, in 2003 and 2005, respectively, and the Ph.D. degree in computer science from the University of York, York, U.K., in 2010.

He is currently a Professor with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao, China. His research interests include marine information processing, remote sensing, and machine learning.

Dr. Ren was a recipient of the K. M. Scott Prize from the University of York in 2011 and the Eduardo Caianiello Best Student Paper Award at the 18th International Conference on Image Analysis and Processing in 2015 as a co-author. He serves as an Associate Editor for journals including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF OCEANIC ENGINEERING, and *Pattern Recognition*.



Weimin Huang (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in radio physics from Wuhan University, Wuhan, China, in 1995, 1997, and 2001, respectively, and the M.Eng. degree in electrical engineering from the Memorial University of Newfoundland, St. John's, NL, Canada, in 2004.

From 2008 to 2010, he was a Design Engineer with Rutter Technologies, St. John's. Since 2010, he has been with the Faculty of Engineering and Applied Science, Memorial University of Newfoundland, where he is currently a Professor. He has authored or coauthored more than 300 refereed research articles. He edited the book *Ocean Remote Sensing Technologies: High Frequency, Marine, and GNSS-Based Radar*. His research interests include mapping oceanic surface parameters via high-frequency ground wave radar, X-band marine radar, synthetic aperture radar, and global navigation satellite systems (GNSSs).

Dr. Huang was a member of the Electrical and Computer Engineering Evaluation Group for Natural Sciences and Engineering Research Council of Canada Discovery Grants from 2018 to 2021. He has been a Technical Program Committee Member. He is an Executive Committee Member of IEEE Oceanic Engineering Society (OES). He was the recipient of a postdoctoral fellowship from the Memorial University of Newfoundland, the Discovery Accelerator Supplements Award from the Natural Sciences and Engineering Research Council of Canada in 2017, and the IEEE Geoscience and Remote Sensing Society Letters Prize Paper Award in 2019, as well as some other teaching and research awards. He was the Co-Chair of the Electrical and Computer Engineering Evaluation Group for the Natural Sciences and Engineering Research Council of Canada Discovery Grants from 2018 to 2021. He is the General Co-Chair and the Technical Program Co-Chair for the IEEE 20th International Symposium on Antenna Technology and Applied Electromagnetics and the IEEE OES 13th Currents, Waves, and Turbulence Measurement Workshop. He is also an Area Editor of IEEE CANADIAN JOURNAL OF ELECTRICAL AND COMPUTER ENGINEERING; an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE JOURNAL OF OCEANIC ENGINEERING, *Remote Sensing, Frontiers in Marine Science, and Frontiers in Remote Sensing*; and a Guest Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and six other journals. He has been a reviewer for more than 130 international journals and a reviewer for many IEEE international conferences, such as the IEEE Radar Conference, International Conference on Communications, IEEE Global Communications Conference, and IEEE International Geoscience and Remote Sensing Symposium.

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